

Problem 1

```
alpha <- 0.02
t <- (-2.373)
n <- 28
pvalue <- 2 * (1 - pt(abs(t), n - 1))
pvalue
```

```
## [1] 0.02502324
```

```
pvalue < alpha
```

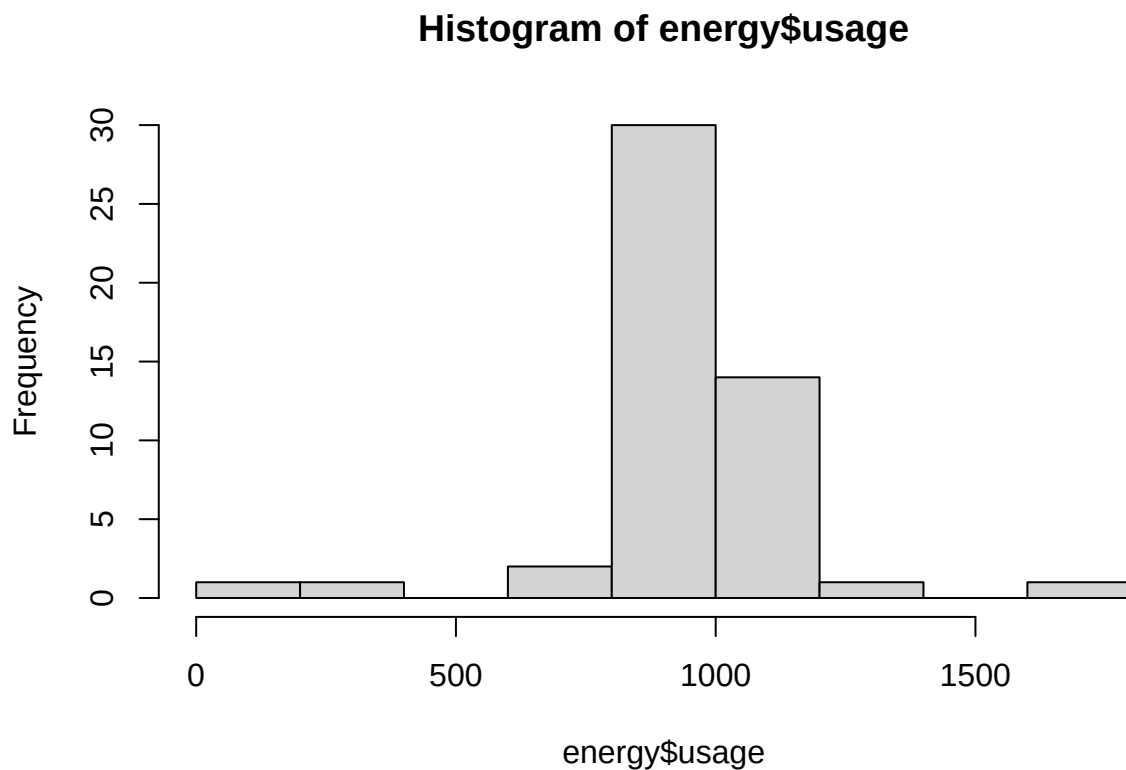
```
## [1] FALSE
```

Problem 2

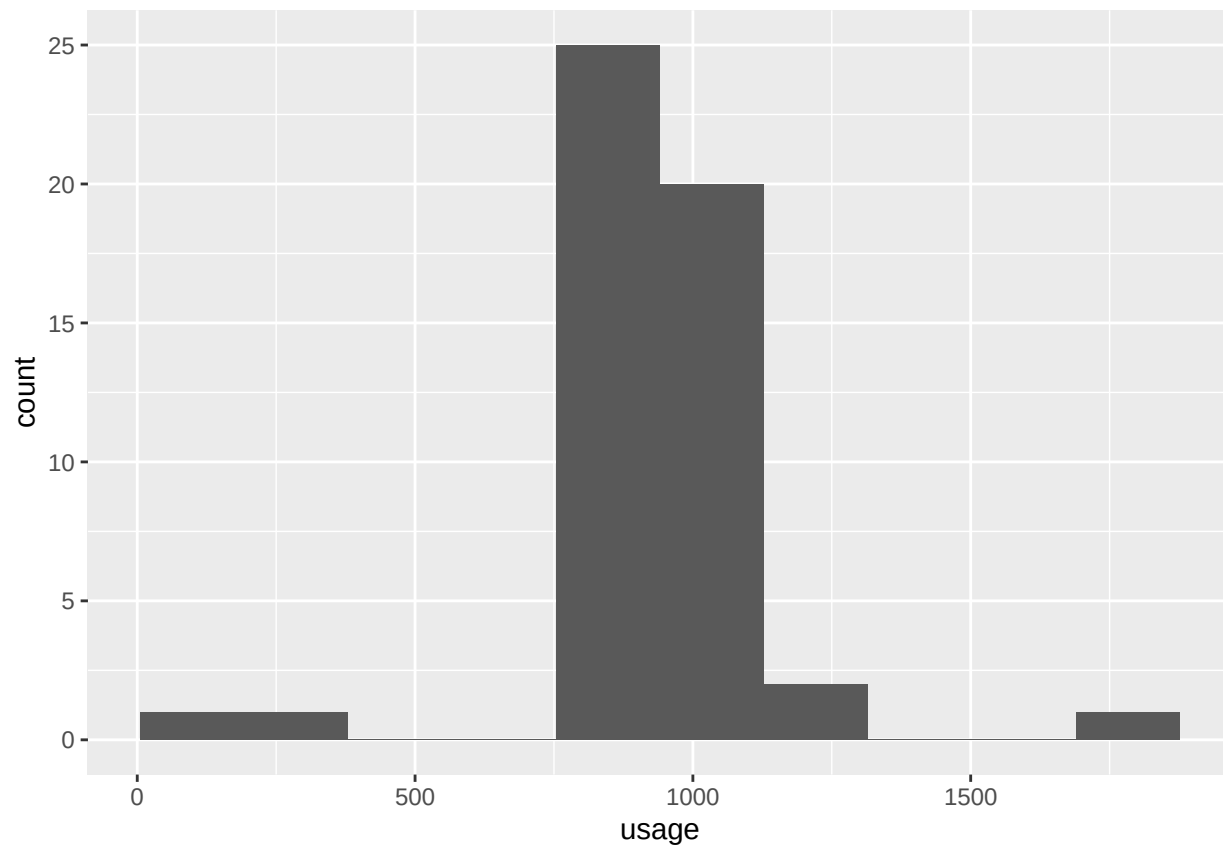
```
energy <- read.csv("../data/energy.csv")
```

A: Plot Data

```
hist(energy$usage)
```



```
library(ggplot2)
ggplot(energy, aes(x = usage)) +
  geom_histogram(bins = 10)
```



B: Relevant Variables

```
# Population
mu <- 909
sigma <- 105

# Sample
alpha <- 0.05
xbar <- mean(energy$usage)
n <- length(energy$usage)
s <- sd(energy$usage)
```

C: Confidence Interval

```
zstar <- 1 - qnorm(1 - alpha)
```

D: Hypothesis Test

$H_0: \mu = 909$

$H_1: \mu > 909$

```
alpha <- 0.05
```

```
stderr <- sigma / sqrt(n)
z <- (xbar - mu) / stderr
z
```

```
## [1] 2.030453
pvalue <- pnorm(z)
pvalue

## [1] 0.9788448
pvalue < alpha

## [1] FALSE
```